

Swarali Kestikar

Arlington, VA

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Education

Virginia Tech

Masters in Computer Science | GPA: 3.83 / 4.00

August 2024 - May 2026
Virginia, USA

MKSSS's Cummins College of Engineering for Women

Bachelor of Technology in Computer Engineering

August 2018 - July 2021
Pune, India

Technical Skills and Certifications

Languages: Java, Python, HTML, CSS, JavaScript (React.js, Next.js), Rust

Databases: SQL, MySQL, Oracle Database, MongoDB

Tools and Technologies: Tableau, Azure, AWS (EC2, Lambda, API Gateway, S3), CI/CD(Jenkins, GitHub Actions), Docker, VS Code, Ansible Tower, Jenkins, JIRA, GIT, Tomcat, Android SDK, Bit Bucket, Spring Boot, REST

Work Experience

Amazon

SDE Summer 2025 Intern

May 2025 – August 2025

Seattle, WA

- **Developed** a **Rust-based** tool to track freshness of **100+ third-party packages** by comparing in-use versions with upstream, ensuring security compliance and up-to-date dependencies.
- **Designed** configurable staleness definitions to systematically detect outdated packages, improving reliability of dependency management.
- **Optimized** package processing pipeline using **Octocrab** for GitHub API integration and **Rayon** for parallelization, reducing execution time from **76 minutes to under 8 minutes (~90% improvement)**.
- Strengthened **automation** and observability in package lifecycle management, reducing manual intervention and improving developer productivity.

Bank of America

Officer; Software Engineer

June 2021 – August 2024

Mumbai, India

- Engineered and improved high-performance **Python-based financial applications** leveraging **Quartz** and **Oracle Database**, improving **transaction efficiency by 50% and system reliability**.
- Developed, tested, and deployed core **enterprise-grade software features**, implementing **unit and integration testing strategies** that improved **system stability by 45%**.
- Enhanced **production resilience** by executing comprehensive **performance and regression testing** in lower environments, reducing **downtime by 35%** during production releases.
- Led a **large-scale process re-engineering initiative**, restructuring end-to-end service processes across **7 upstream systems**, reducing **paperwork by 70%** and enhancing **cross-team collaboration**.
- Orchestrated **critical server migrations**, applying **fault-tolerant architectures** and **disaster recovery strategies**, cutting **downtime by 30%** while ensuring **seamless operational continuity**.
- Automated key **operational workflows** using **Python scripting**, eliminating **40% of manual effort**, improving **system efficiency**, and enabling **faster issue resolution**.
- Implemented **documentation automation tools**, streamlining **internal knowledge transfer**, reducing **onboarding time by 25%**, and fostering **team-wide efficiency**.

Academic Projects

Movie Ticket Booking System | React, Selenium, AWS SDK/EC2/Lambda/DynamoDB, Speech Synthesis

- Implemented **Python script** to **crawl and scrape 30+ websites**, gather **movie data**, and create **entries in the database**.
- Built **serverless backend** in **Python** using **AWS SDK, API Gateway, and Lambda functions** with appropriate triggers to handle **movie information in DynamoDB** and corresponding **movie trailers/images in S3 buckets**.
- **Launched React frontend** build on **Unix-based AWS EC2 instance**.

Bookstore Web Application | React JS, Typescript, JavaScript Node.js, Express.js, MySQL, HTML, CSS

- Created a **full-stack React-based e-commerce platform** with **Node.js & Express**, implementing **secure transactions, inventory management**, and optimized **UI/UX** for seamless book purchases.
- Integrated **RESTful APIs with MySQL** and implemented **secure payment processing**, ensuring **high performance, scalability, and PCI compliance**.
- Deployed the application using **Docker** and **CI/CD pipelines (GitHub Actions)**, automating testing and deployment for **enhanced reliability and efficiency**.

Awards & Honors

Gold and Silver Awards: Received dual **Global Recognition Awards** for **demonstrated commitment, unwavering dedication, and quality assurance efforts**, contributing to the **success of the project**.